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5 Things Fleet Owners Should Check Before Switching from Lead-Acid to Lithium Batteries

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Battery Solutions for AMR, AGV, UGV & Mobile Robotics | Custom Li-ion Battery Packs | OEM/ODM | BMS & CAN...

August 9, 2026

In recent years, more and more fleet operators are considering replacing traditional lead-acid batteries with lithium batteries.

The reasons are clear:

- Longer service life
- Lower maintenance requirements
- Higher energy efficiency
- Better performance in demanding applications

However, replacing a lead-acid battery with a lithium battery is not simply a “drop-in replacement”.

Especially for heavy-duty vehicles, commercial fleets, and industrial equipment, the battery is not only an energy storage device. It is part of the entire electrical system.

Before making the switch, fleet owners should carefully evaluate the following five points.

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1. Starting Performance: Capacity Is Not the Only Key Factor

Many people compare batteries based only on Ah capacity.

But for heavy-duty vehicles, starting performance is often more important than capacity.

A truck engine requires a very high current during startup, especially in cold weather conditions. A battery that has enough capacity but cannot provide sufficient peak current may cause:

- Slow engine cranking
- Voltage drop
- Starting failure
- Stress on vehicle electronics

Therefore, when evaluating lithium batteries for commercial vehicles, fleet owners should pay attention to:

- Peak discharge current
- Voltage stability under high load
- Internal resistance
- Low-temperature starting performance

A reliable battery is not just about storing more energy. It must deliver power when the vehicle needs it most.

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2. Charging Compatibility: The Battery Must Work With the Vehicle System

One common misunderstanding is:

"Same voltage means the lithium battery can directly replace the lead-acid battery."

In reality, charging systems are different.

Heavy-duty trucks usually use alternators designed for lead-acid batteries. During deep discharge, the alternator can provide a high charging current, which may exceed the safe charging limit of a lithium battery.

A professional lithium battery solution should consider:

- Charge current limitation
- BMS protection strategy
- Communication with the vehicle system
- Protection against abnormal voltage conditions

The goal is not only to charge the battery faster, but to ensure long-term safety and reliability.

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3. Temperature Management: Cold Weather Performance Cannot Be Ignored

Temperature is one of the biggest challenges for lithium batteries.

In cold environments, lithium batteries can still discharge well, but charging below 0°C requires special protection to avoid lithium plating and potential battery damage.

For applications operating in cold regions, fleet owners should check:

- Does the battery have a heating system?
- How does the BMS manage low-temperature charging?
- How long does pre-heating take?
- What is the available capacity at low temperatures?

A battery designed for a warm climate may not perform well in harsh winter conditions.



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4. BMS Capability: The Intelligence Behind the Battery

The battery cells are important, but the Battery Management System (BMS) determines how safely and efficiently the battery operates.

A reliable BMS should monitor and protect:

- Cell voltage
- Temperature
- Charging and discharging current
- Cell balancing
- Overcharge and over-discharge protection
- Short-circuit protection

For commercial fleets, more advanced functions become increasingly important:

- CAN / J1939 communication
- Remote monitoring
- Fault diagnosis
- Battery health tracking

When hundreds or thousands of vehicles are operating, fleet managers need visibility into battery conditions before problems happen.

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5. Long-Term Reliability: Think Beyond the First Purchase Price

The lowest battery price does not always mean the lowest operating cost.

For fleet applications, the real value comes from:

- Longer service life
- Reduced maintenance
- Less downtime
- Better operational efficiency

Before selecting a lithium battery supplier, fleet owners should evaluate:

- Battery testing standards
- Production quality control
- Warranty conditions
- After-sales support
- Experience with similar applications

A battery installed in a commercial vehicle may need to work reliably for years in vibration, temperature changes, and daily operation.



The transition from lead-acid to lithium batteries is not only a technology upgrade. It is a system improvement.

The right lithium battery solution should match the vehicle, the operating environment, and the real working conditions.

A professional battery supplier should not only provide cells and a battery pack. More importantly, they should understand the challenges behind the application and help customers build a safer, more reliable energy system.

Because in commercial applications, the best battery is not always the one with the highest capacity or lowest price.

It is the one that keeps working when the vehicle needs it most.

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